

Why Enterprises Pay the Frontier Premium

REV 2

September 2026 update · five mechanisms behind the closed-vs-open procurement decision · modelled economics

STRUCTURED ABSTRACT

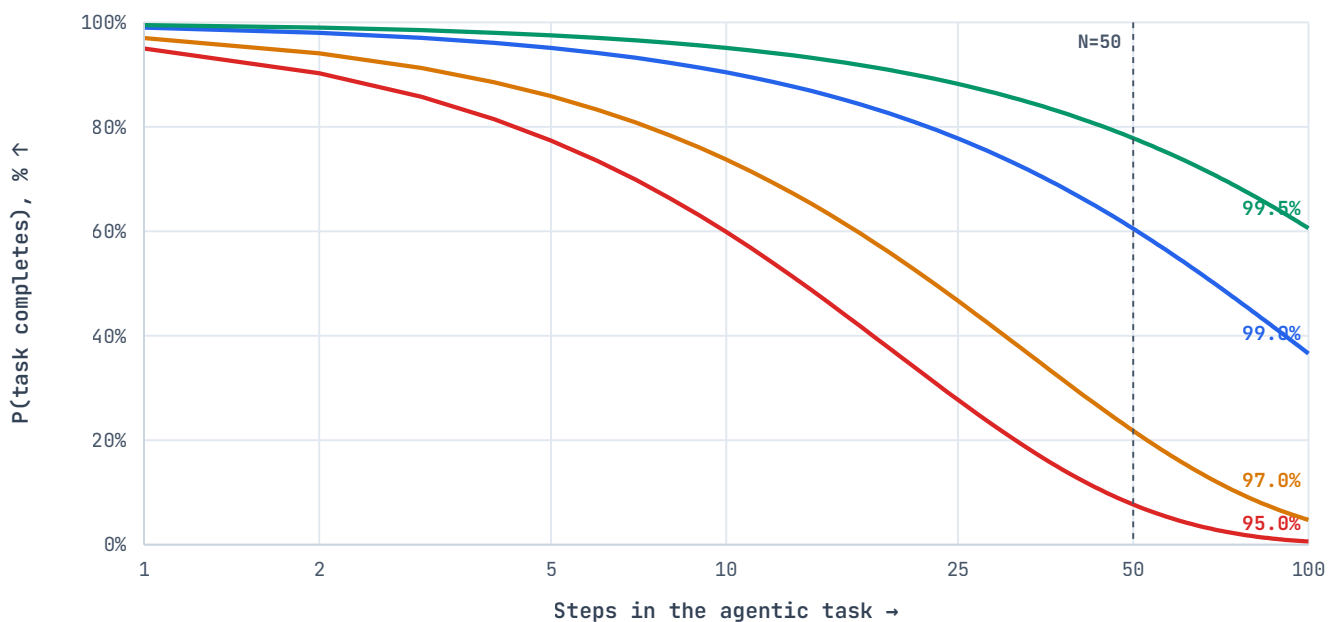
QUESTION	Why do large enterprises pay a substantial premium for frontier proprietary models when open weights reach roughly 87% of composite benchmark capability at a fraction of the token price?
FINDING	Token price is the wrong denominator. Cost per unit of output is $(1 + f) / m$: model spend enters additively while throughput divides. At realistic spend the frontier needs only a ~4% throughput edge to break even, a bar that per-step reliability compounding clears on any workload of length.
METHOD	Derived arithmetic (p^N compounding; the unit-cost identity and its isoclines) combined with parameterised models for self-hosting TCO and threshold-gated value. Benchmark and reliability figures are third-party reported.
SCOPE	Applies to knowledge-work and agentic software deployment at enterprise scale. Does not address consumer products, on-device inference, or research settings.

Token price is the wrong denominator. Cost per unit of output is $(1 + f) / m$ – model spend is additive, throughput is a divisor. At a revised **2.4–5.8% of loaded labour**, the frontier needs only a **4.0% throughput edge** over open weights to break even – a bar that p^N compounding clears on any workload of length. The genuine overpayment isn't choosing proprietary; it's **routing everything to a flagship max-effort tier**.

1 · Agentic error compounding

Task completion probability vs horizon · $P(\text{success}) \approx p^N$

The largest single driver, and the one invisible on single-turn benchmarks. A two-point per-step reliability delta rounds away in a composite index and is **decisive** by step 50.



PER-STEP RELIABILITY	10 STEPS	25 STEPS	50 STEPS	100 STEPS
● 95.0%	59.9%	27.7%	7.7%	0.6%
● 97.0%	73.7%	46.7%	21.8%	4.8%
● 99.0%	90.4%	77.8%	60.5%	36.6%
● 99.5%	95.1%	88.2%	77.8%	60.6%

WHY THIS DOMINATES

Moving per-step reliability from **97% to 99%** is a **2.8×** swing in completion at **N=50** and **7.4×** at **N=100** – the boundary between "the agent ships the PR" and "an engineer babysits and re-drives." Read against driver 2: an edge this size clears the 4.0% break-even bar many times over.

2 · The labour denominator

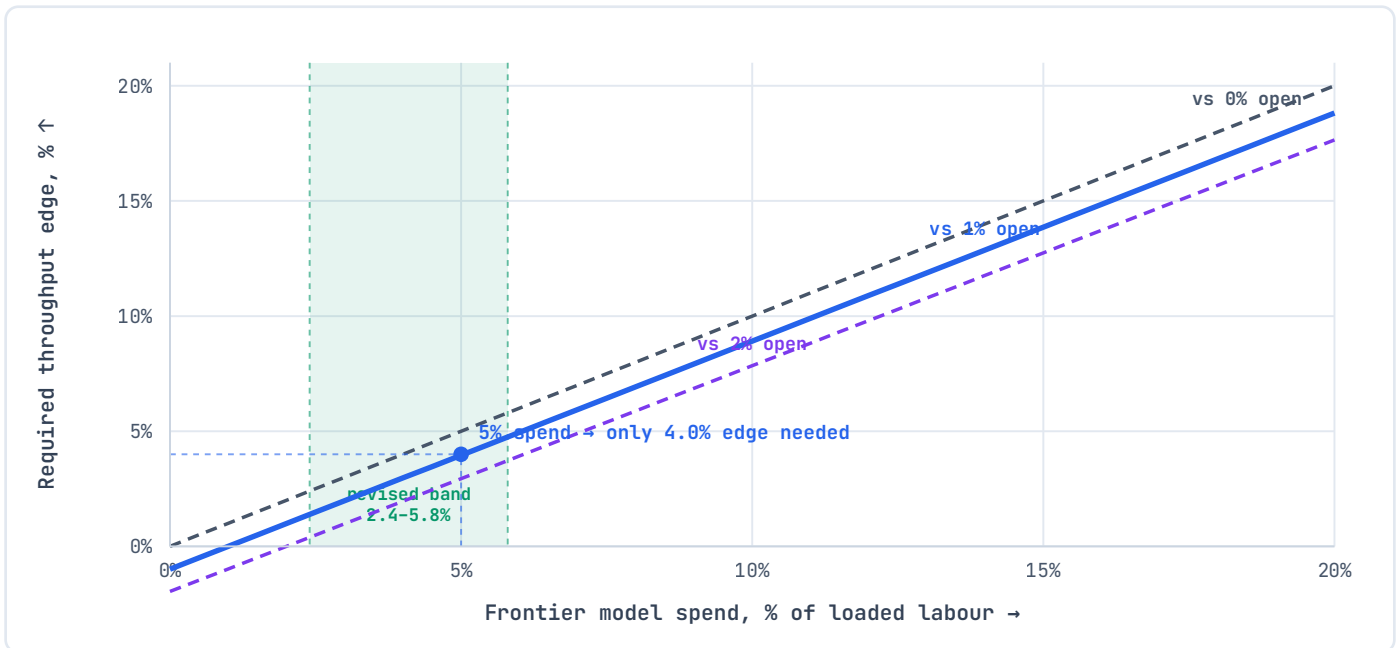
REVISED

REFRAMED

Unit cost = $(1 + f) / m \cdot f$ = spend as a fraction of loaded labour, m = throughput multiplier

f is additive; m is a divisor. A small numerator penalty buys a large denominator gain, which is why the frontier wins on unit cost while spending several times the tokens.

SCENARIO	F · SPEND	M · THROUGHPUT	UNIT COST	VS BASELINE
● Baseline · no tooling	0%	1.00×	1.000	—
● Open weights	1%	1.20×	0.842	-15.8%
● Frontier	5%	1.45×	0.724	-27.6%
● Frontier · heavy agentic	10%	1.55×	0.710	-29.0%



BREAK-EVEN AT 5% SPEND

+4.0% edge

BREAK-EVEN AT 10% SPEND

+8.9% edge

BREAK-EVEN AT 15% SPEND

+13.9% edge

FRONTIER UNIT COST

0.724 vs 0.842 open

REVISED SPEND BAND – THE EARLIER 1-2% WAS ANCHORED TOO LOW

A saturated agentic engineer running parallel sessions lands nearer **\$500-\$1,200/seat/month**, or **2.4-5.8% of \$250k loaded labour**, with a fat right tail (MED – inferred from agentic usage patterns, not a surveyed figure). Raising f does not weaken the case: at 5% frontier spend against 1% open weights, break-even is a **4.0% throughput edge**. Even at 15% spend it is only 13.9%. The token-price argument was never load-bearing.

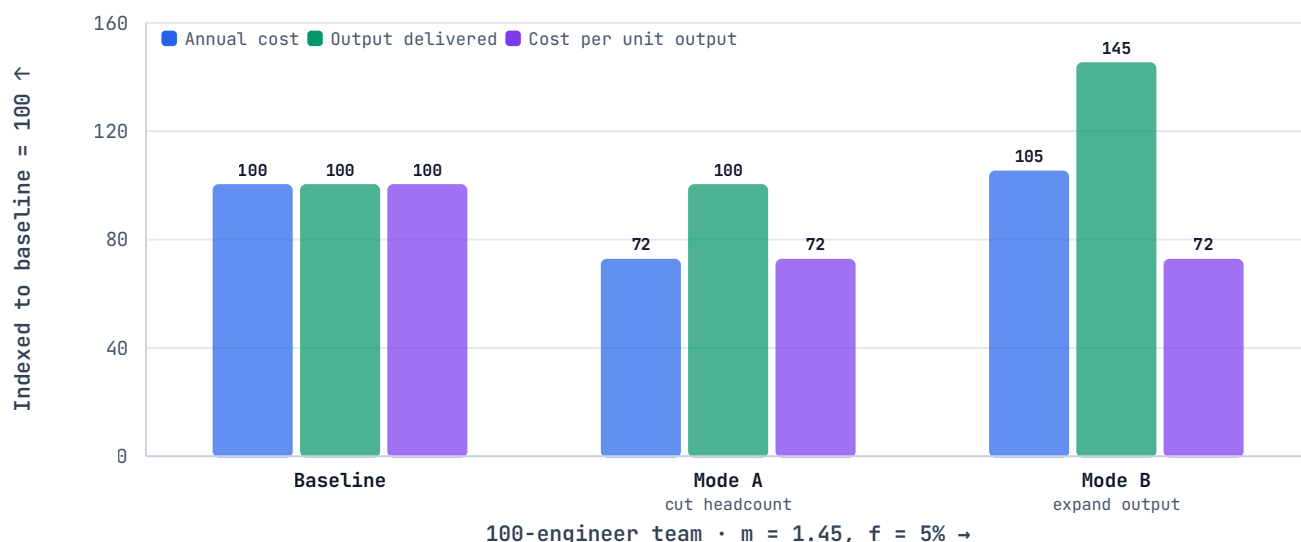
JEVONS DRIFT – AND WHERE THE FRAME BREAKS

As inference gets cheaper per unit of work, consumption rises faster than unit cost falls, so **total spend goes up**. The 2.4-5.8% figure will drift upward, and that drift is a health signal rather than a cost problem. Past some point "tooling as a percentage of labour" stops being the right accounting frame entirely: compute becomes a **factor of production alongside labour** rather than an overhead line charged against it. That is when the procurement conversation changes shape.

3 • Substitution modes NEW

Two ways to bank the same multiplier • identical on unit cost, different on total value

A throughput multiplier can be taken as fewer engineers for the same output, or the same engineers for more output. Illustration: a 100-engineer team at $m = 1.45$, $f = 5\%$.



MODE A

Reduce headcount

Value is **capped at the labour line** – you recover at most the salaries eliminated, and the gain terminates. Second-order trap: shrinking the denominator makes model spend rise as a percentage of labour mechanically, so the ratio you are managing starts moving on you. **Forced when demand is the binding constraint.**

MODE B

Hold headcount, expand output

Value is bounded by **opportunity set, not by cost**. Unbounded upside where a project backlog exists – additional revenue-generating work rather than a one-time cost recovery. **Available when engineering capacity is the constraint.**

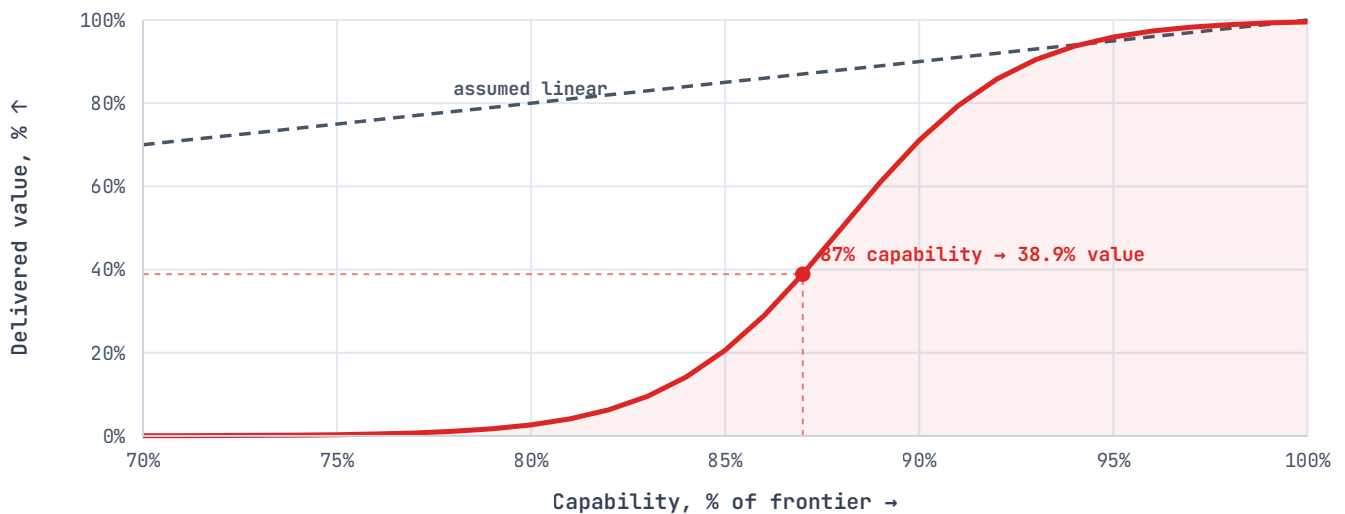
THE CHOICE IS A DIAGNOSTIC, NOT A PREFERENCE

Both modes land at **72.4 on cost per unit of output** – the arithmetic cannot distinguish them. What separates them is which constraint actually binds. Most software organisations claim engineering capacity is their limit, often accurately; firms that reflexively take Mode A are usually revealing that their real constraint was **never headcount**. That is a strategy disclosure, not a procurement decision.

4 • Threshold-gated value

Delivered value vs capability • the composite-average trap, priced

"Open weights reach ~87% of proprietary" assumes linear value transfer. On work gated by a pass threshold **delivered value is sigmoid**, and the last few index points sit on the knee.



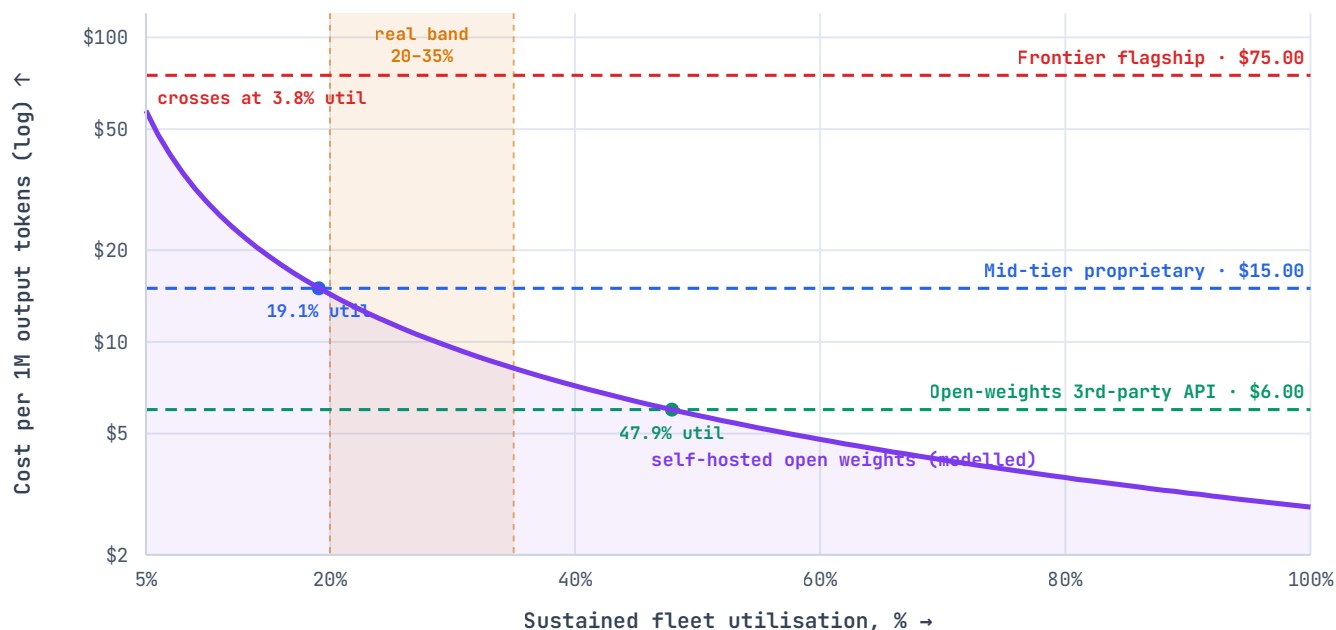
READING IT

On this curve **87% of frontier capability delivers ~38.9% of the value**. The benchmark isn't wrong – a mean over ten evals is simply the wrong functional form for threshold-gated work. The easy 80% of tasks were automatable two model generations ago; the residue is exactly what sits on the knee. Illustrative – LOW confidence on curvature, HIGH on direction.

5 • Self-hosting total cost of ownership

Effective cost per 1M output tokens vs fleet utilisation • $4 \times 8 \times H200 + 3 \text{ ops FTE}$

Open weights are free; **serving them is not**. Fixed cost amortises over tokens actually produced, so unit cost is governed by utilisation – and enterprise diurnal load runs **5-10× peak-to-trough**.



BREAK-EVEN VS OPEN-WEIGHTS
3RD-PARTY API

47.9% util

BREAK-EVEN VS MID-TIER
PROPRIETARY

19.1% util

BREAK-EVEN VS FRONTIER
FLAGSHIP

3.8% util

FIXED COST MODELLED

\$1.81M /yr

MODEL ASSUMPTIONS – LOAD-BEARING, TAGGED

Infra **\$911k/yr** (4 nodes × \$26/hr rented – MED) plus ops **\$900k/yr** (3 FTE × \$300k loaded – MED) = **\$1.81M/yr fixed**. Peak capacity 5,000 output tok/s per node for a ~100B-class MoE under heavy batching – LOW confidence; this is the swing variable. Halve throughput and every crossover doubles. The qualitative result survives that: self-hosting beats a flagship price almost immediately and a cheap open-weights API only at high sustained load.

Net read

Nine drivers, with basis and confidence

DRIVER	MECHANISM	BASIS	CONF
Agentic error compounding	97%~99% per-step = 2.8× completion at N=50	Derived (p^N)	HIGH
Labour denominator	f is additive, m is a divisor – break-even edge only ~4%	Derived	HIGH
Substitution symmetry	Cut headcount and expand output share identical unit cost	Derived	HIGH
Threshold-gated value	87% composite capability → ~39% delivered value	Modelled	MED
Calibration inversion	Opus 4.7 hallucination 36% vs GPT-5.5 xhigh 86%	AA, Jun 2026	HIGH
Self-host TCO	Break-even vs cheap API needs ≈48% sustained utilisation	Modelled	LOW
Indemnity & compliance	IP indemnity, SOC 2 / HIPAA / FedRAMP, residency	Contractual	HIGH
Provenance & sanctions	Training-data lineage, backdoor risk, Entity List exposure	Qualitative	MED
Harness, not checkpoint	Caching, batch, tool-call reliability, agentic scaffolds	Qualitative	HIGH

LANE 1 • OPEN WEIGHTS

Bounded, high-volume, private

Classification, extraction, routing, embedding. Error-tolerant, short-horizon, sustained enough to clear the utilisation bar. **The open-weights case holds cleanly here.**

LANE 2 • MID-TIER PROPRIETARY

General reasoning, good-enough

Where the **dominated-flagship trap** lives. Cost-tuned proprietary out-values open weights on both axes – same index, a fifth the cost.

LANE 3 • FRONTIER

Long-horizon, calibration-critical

Agentic chains, must-not-hallucinate, threshold-gated. p^N and the sigmoid knee are both load-bearing. **Token cost is the small line item.**

NET

Enterprises don't pay the premium because they're bad at arithmetic – they pay it because the arithmetic they're doing has a different denominator, and because f enters additively while m divides. Where open weights genuinely win is a real and sizeable quadrant, but it's **bounded work at sustained load**, not general reasoning. The overpayment that actually exists is **failure to segment traffic**, and it shows up inside the proprietary cohort as often as across the closed/open line.

Changes in rev 2. Spend band revised upward from 1-2% to 2.4-5.8% of loaded labour; driver 2 reframed from a break-even bar chart to the $(1 + f) / m$ unit-cost identity and its isoclines; substitution modes added as a distinct driver; Jevons drift noted.

Sources & confidence. Reliability compounding and the unit-cost identity are derived arithmetic – HIGH. Hallucination rates (Opus 4.7 36%, GPT-5.5 xhigh 86%, Gemini 3.1 Pro 50%) and index/cost figures from Artificial Analysis Intelligence Index v4.0, live June 9 2026 – HIGH. Throughput multipliers ($m = 1.20 / 1.45 / 1.55$) are **illustrative parameters, not measurements** – LOW; they set the size of the gain, not the direction, and break-even is independent of them. Spend band MED. Loaded labour \$250k is a representative US senior-engineer figure – MED. Self-host TCO is a model, not a measurement: node rental \$26/hr and \$300k loaded FTE are market estimates (MED); 5,000 output tok/s per node is an assumption (LOW) and dominates that result. The threshold curve is illustrative – LOW on curvature, HIGH on direction.

Not included: project-level discounts, committed-use pricing, and enterprise volume agreements materially compress the frontier premium and are absent from every figure here. Reference API prices are per 1M output tokens and exclude input, caching, and the verbosity tax on output-heavy models.

Notes & clarifications

Everything that is a choice rather than a measurement, stated plainly

Change log

Labour denominator · spend band REVISED

Model spend raised from 1-2% to 2.4-5.8% of loaded labour (\$500-1,200/seat/month). The earlier band was anchored on light interactive use and understated saturated agentic sessions. The revision strengthens rather than weakens the conclusion – break-even rises only to 4.0%.

Labour denominator · presentation REFRAMED

Replaced the break-even bar chart with the $(1 + f) / m$ unit-cost identity and its isoclines. The bar chart answered "what lift repays the spend" against an absolute baseline; the isoclines answer the decision-relevant question, which is the lift required relative to the open-weights alternative.

Substitution modes NEW

Added as a distinct driver. Headcount reduction and output expansion are algebraically identical on cost per unit of output and differ only in where value is booked and whether it is bounded.

Jevons drift NEW

Noted that total spend rises as unit cost falls, and that percentage-of-labour eventually stops being the correct accounting frame.

Prior-revision corrections CARRIED

Corrections from the previous edition remain in force: the provenance-launders calibration attribution and the invalid hallucination-rate denominator. The calibration-inversion driver remains withdrawn rather than reversed.

Notes

1 Throughput multipliers are parameters, not measurements LOW

$m = 1.20 / 1.45 / 1.55$ are chosen to illustrate the shape of the unit-cost relation. They set the size of the gain, not its direction. Break-even is independent of m entirely – it is $(1 + f_{\text{frontier}}) / (1 + f_{\text{open}})$ – so the 4.0% figure survives any revision to them. First-party measured throughput would rank above these and should replace them.

2 p^N is an upper bound on decay, not a forecast HIGH

The independence assumption overstates compounding: real agentic failures correlate, and recovery, retry, and human checkpoints truncate the chain. The curves describe the shape of the effect and the direction of the reliability premium. Specific completion rates are illustrative.

3 Spend band is inferred, not surveyed MED

\$500-1,200/seat/month is reasoned from agentic usage patterns rather than measured across a population. The right tail is fat: parallel-session users can exceed it materially.

4 Self-host TCO is dominated by one assumption LOW

5,000 sustained output tok/s per node for a ~100B-class MoE governs every crossover in driver 5. Halve it and each crossover doubles. Node rental and loaded FTE cost are market estimates. The qualitative ordering survives; the specific percentages do not.

5 Threshold curve is illustrative on curvature MED

The sigmoid's steepness and midpoint are chosen, not fitted. The direction – that value transfer is convex rather than linear on gated work – is the load-bearing claim.

6 No composite averaging across heterogeneous rows HIGH

The 87% figure is quoted only as the thing being critiqued. It is a mean over ten evals and is not used here as an input to any calculation.

7 Reference prices are output-only MED

API prices exclude input tokens, cache reads, and the verbosity tax on output-heavy models. Folding verbosity in shifts effective frontier cost upward and moves the self-hosting crossovers left.

8 Commercial terms are excluded HIGH

Project-level discounts, committed-use pricing, and enterprise volume agreements materially compress the frontier premium and appear in no figure here. Every stated premium is therefore an upper bound on what a negotiating enterprise actually pays.